

# Necessity Entrepreneurs,<sup>\*</sup>

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## Abstract

TO BE COMPLETED.

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# 1 Introduction

Entrepreneurship is often treated as a driver of innovation, productivity, and growth. However, not all entrepreneurship has the same economic role. In many settings, a substantial share of entry is necessity-driven rather than opportunity-driven. This distinction matters because necessity entry can reflect weak labor-market outside options rather than comparative advantage in managing production.

Empirical work typically separates necessity and opportunity entrepreneurship using survey responses or prior labor-market status. Both approaches are useful, but each has limits. Self-reported motives are subjective, while classifications based only on previous employment status can misclassify agents whose timing is strategic or constrained by insurance and search frictions. A worker who expects separation may enter early and look “opportunity” in the data; an unemployed household can enter temporarily while searching and look “necessity” even when underlying talent is high.

This paper provides a model-based definition that maps directly into counterfactual behavior. We define opportunity entrepreneurs as households that choose entrepreneurship regardless of whether they enter the period employed or non-employed. We define necessity entrepreneurs as households that choose entrepreneurship only when they lack an employment relationship. The model combines heterogeneous managerial talent, human-capital heterogeneity, search frictions, and social insurance, allowing us to quantify how labor-market risk and policy jointly shape the composition of entrepreneurship.

Our contribution is threefold. First, we provide a structural framework in which necessity entrepreneurship emerges endogenously from occupational choice under incomplete insurance. Second, we provide a quantitative decomposition of how unemployment risk and unemployment insurance affect entry margins and firm scale across education groups. Third, we reinterpret necessity entry as a misallocation channel: low outside options can push households into entrepreneurship even when their productivity is better used in paid work.

The rest of the paper proceeds as follows. Section 2 discusses related literature. Section 3 presents the economic environment. Section 4 characterizes household behavior and equilibrium. Section 5 discusses calibration. Section 6 presents baseline quantitative results. Section 7 studies policy counterfactuals, and Section 8 concludes.

# 2 Literature Review

This paper contributes at the intersection of four literatures: occupational choice, entrepreneurship under financial frictions, labor-market search, and heterogeneity in entrepreneurial motives.

The first strand is occupational-choice theory with heterogeneous ability and outside options. Lucas (1978) [CHECK] provides the canonical span-of-control benchmark. Evans and Jovanovic (1989) [CHECK] and Hurst and Lusardi (2004) [CHECK] show that liquidity constraints shape

who can enter entrepreneurship. Cagetti and De Nardi (2006) [CHECK] then quantifies how entrepreneurial frictions and wealth dynamics interact in general equilibrium.

The second strand studies entrepreneurship and firm dynamics under financial constraints over the cycle. Buera (2009) [CHECK] shows that credit frictions affect both entry and post-entry scale. Poschke (2013b) [CHECK] is especially close to our setup: labor-market prospects are central for occupational choice into entrepreneurship. Sedlacek and Sterk (2017) [CHECK] further show that startup cohorts differ in growth potential over the business cycle.

The third strand is the search-and-matching labor literature. Mortensen and Pissarides (1994) [CHECK] and Petrongolo and Pissarides (2001) [CHECK] provide the benchmark framework for job-finding and separation dynamics that discipline our transition structure.

The fourth strand emphasizes entrepreneurial motives and composition. Fairlie and Fossen (2019) [CHECK] distinguish opportunity and necessity entrepreneurship in the data, while Hurst and Pugsley (2011) [CHECK] show that most small-business activity is not primarily innovation-driven. Poschke (2013a) [CHECK] provides direct evidence that necessity-based entry is quantitatively important in cross section. These findings motivate our focus on composition, not only the aggregate entrepreneurship rate.

Relative to existing work, our contribution is to integrate these channels in a single heterogeneous-agent framework with explicit state-contingent occupational choice. This allows us to quantify how unemployment risk and social insurance reshape the mix of necessity and opportunity entrepreneurship, firm scale, and aggregate outcomes.

### 3 The Model: Economic Environment

Consider a Lucas (1978) span of control model, where individuals differ in the level of human capital,  $h$ . As in Gomez and Kuhen (2017), individuals are educated to either the primary ( $P$ ), secondary ( $S$ ) or tertiary level ( $T$ ). The level of education means that a tertiary educated individuals have a higher draw of human capital, on average, than secondary educated individuals, who in turn have a higher draw on average than primary educated individuals. This human capital draw is positively correlated with managerial ability  $x$  as well as labor productivity  $z$ .

At the start of the period agents enter as either unemployed ( $i_w = 0$ ), employed ( $i_w = 1$ ), or an entrepreneur ( $i_w = 2$ ). Individuals face a choice of whether to become an entrepreneur, to continue working or to search for work. However, the value of employment is different depending on whether an agent enters the period as employed ( $i_w = 1$ ) or not. Although there is a risk that an employee may become separated from a firm with probability  $\rho$ , this is less risky than searching for work which has a  $p(\theta)$  probability of obtaining employment. Consequently, the expected value of employment is higher for those already in employment than those who are not. Additionally, the probability of separation is lower for individuals with higher levels of human capital and they are able to command a higher wage due to higher levels of productivity. So the

value of employment, holding other things equal, is higher for those with higher human capital as well.

The innovation of this model is that the entrepreneurial decision results in two types of entrepreneurs emerge, opportunity-based entrepreneurs and necessity-based entrepreneurs. We define opportunity-based entrepreneurs are those whose value of entrepreneurship is higher, even if they had employment. Necessity-based entrepreneurs are those who would not have become entrepreneurs, if they had employment.

**Timing:** The timing of events can be summarized below. Occupational choice happens at the start of time  $t$ . Those choosing to become entrepreneurs run projects and hire workers on on rolling contracts (i.e. contracts, will be reassessed each period). Those who were previously employed within a project can decide to carry on working but carry a risk of separation with probability  $\rho$ . If separation occurs these workers will claim unemployed benefits and be unable to look for work until the following period. After separation occurs, vacancies are posted and who were unemployed in the previous period are matched with firms. Those that are successful and match with with probability  $p(\theta)$  go on to earn a wage. Whereas those that are unsuccessful will claim unemployment benefits and be unable to search again until the following period.

Unlike the search and matching literature there is no wage bargaining between workers and the firm at the end of matching. Instead the wage pins down the occupational choice decision which determines those wanting to become entrepreneurs and those who will search for work. Crucially, the worker only finds out whether they enter production and earn a wage after their occupational choice decision.

Figure 1: Timing for Matched Workers

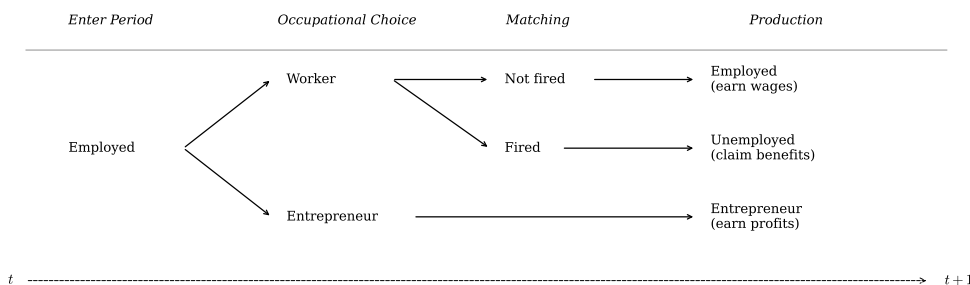
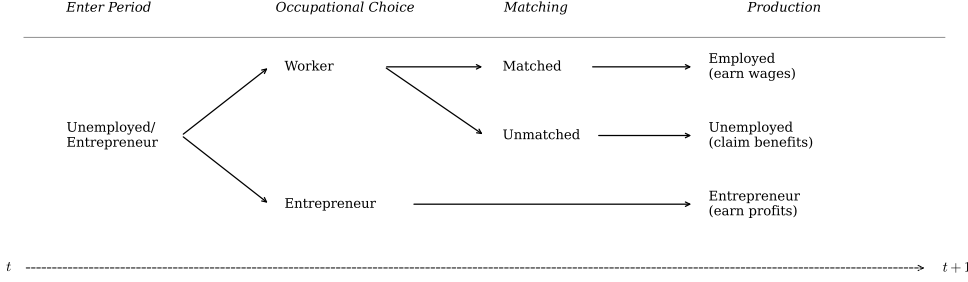


Figure 2: Timing for Unmatched Workers



### 3.1 Preferences, endowments and technology

**Preferences:** Consumption by an agent in period  $t$  is  $c_t$ , with utility given by  $U(c_t)$ .

**Endowments:** Agents are endowed with an initial capital asset,  $a_0$ , which can be used as an input in production. Each individual is endowed with human capital shock  $h^i$ . Human capital  $h$  is positively correlated with managerial talent  $x$ , labor productivity  $z$  and separation rate  $\rho$ .

**Production:** Firms use labor  $n$  and capital  $k$  to produce a single consumption good,  $y$ . Capital depreciates at a constant rate of  $\delta$ . Managers can operate only one project. The functional form of the production function is:

$$y = xk^\alpha(zn)^\gamma \quad \text{where } \alpha, \gamma > 0. \quad (1)$$

Managerial talent  $x$  is determined by the Pareto distribution  $F(x) = \left[1 + \frac{x-\varsigma}{\xi}\right]^{-\vartheta}$  where  $\vartheta$  is the shape parameter,  $\varsigma$  the location parameter and  $\xi$  is the scale parameter.

**Factor remuneration:** Firms rent capital at the common market rate  $r$ .

**Government:** The government runs a balanced budget each period and provides unemployment insurance which is financed through lump sum taxation  $\tau$ , which is shared by the employee  $(1 - \psi)\tau$  and employer  $\psi\tau$ . This program guarantees each household an unemployment benefit of  $b$ .

### 3.2 The Firm's problem

The firm's problem is:

$$\max_{n,k} Xk^\alpha n^\gamma - (1 + \tau\psi)wzn - rk - \kappa \frac{\rho zn}{q(\theta_t)} \quad (2)$$

employees must pay a share of unemployment insurance  $\tau$  to all workers who produce output. Firms have to pay search costs  $\kappa$  for all vacancies posted. The number of vacancies posted is designed to replace any lost workers,  $\rho zn$ . This, however, is affected by the firm-hiring rate  $q(\theta_t) = \bar{m} \left( \frac{v_t}{u_t} \right)^{-\mu}$ , where aggregate vacancies  $v_t$  is simply the aggregate level of all those who have separate from firms  $\sum \rho zn$  within the period and  $u_t$  is the total unemployed looking for work within that period (see appendix).

When the market is tight and the hiring rate is low, firms have to search more and thus face higher search costs in order to replace lost workers. This is because  $0 < q(\theta_t) < 1$ , so a lower  $q(\theta_t)$  increases  $\frac{\rho zn}{q(\theta_t)}$ . If the firm lose 10 workers, they want to hire 10 workers to replace them. But if the hiring rate,  $q(\theta_t)$ , is 50% then they will face double the amount of search costs than if  $q(\theta_t)$  were 100%. The upshot of this is that when markets are tight, the size of the desired firm size  $n$  decreases ex ante, as the cost of replacing workers is high and so entrepreneurs would prefer to run smaller firms.

To avoid issues of directed search, we assume that employees are randomly matched with firms (as in Chivers et al., 2015) and each firm employees a distribution of workers across the human capital spectrum such that each firm employs the market average.

## 4 Optimal behavior and equilibrium

In any time period  $t$  agents are distinguished by  $(h, a, x, i, z)$ . The timing of the economy is given as follows.

1. Households enter each new period with assets  $a$  and an employment status  $i_w$
2. Idiosyncratic shocks  $h$  is drawn which determines  $x$  (managerial ability) and  $z$  (labor productivity),  $\rho$  separation rate conditional on education level,  $P, S$  or  $T$ .
3. Households make an occupation decision: entrepreneur ( $\mathcal{I}_e = 1$ ) or worker ( $\mathcal{I}_e = 0$ ).
4. Capital and labor markets clear. A fraction  $\rho$  of employees separate from firms and  $p(\theta)$  are hired to replace them
5. Households choose: consumption ( $c$ ), borrowing/saving ( $a'$ )

### 4.1 Firm manager

Firms are distinguished by their productivity realization  $x$ . Agents with sufficient ability to become managers choose the level of capital and the number of employees to maximize profit subject to a technological constraint. In order to simplify the exposition, first consider the problem of a manager with talent  $x^i$  for a given level of capital  $k$  (i.e., labor input choice only):

$$\max_{zn} xk^\alpha (zn)^\gamma - zw(1 + \psi\tau)n - rk - \kappa \frac{\rho zn}{q(\theta)} \quad (3)$$

The first order conditions are:

$$\gamma x k^\alpha (zn)^{\gamma-1} = w(1 + \psi\tau) + \frac{\rho\kappa}{q(\theta)} \quad (4)$$

$$n^*(k, x, \tilde{w}) = \left[ \frac{\gamma x k^\alpha}{w(1 + \psi\tau) + \frac{\rho\kappa}{q(\theta)}} \right]^{\frac{1}{1-\gamma}} \quad (5)$$

Substituting (5) into (3) yields the manager's profit function for a given level of capital:

$$y(k, x, \tilde{w}) = x k^\alpha (zn^*)^\gamma - zn^* w(1 + \psi\tau) - \kappa \frac{\rho z n^*}{q(\theta)} \quad (6)$$

#### 4.1.1 Capital

Initially we assume capital markets function perfectly.

### 4.2 Entrepreneurs

Entrepreneurs maximize expected discounted utility of consumption

$$\max_{\{c_t, a_{t+1}\}} \mathbb{E} \sum_{t=0}^{\infty} \beta^t U(c_t)$$

subject to the following budget constraint:

$$c + a' \leq (1 + r - \delta)a + \nu - \tau_\pi \quad (7)$$

$$a' \geq -\bar{a}$$

### 4.3 Employed

The workforce (previous periods employed who choose to reapply for work) maximize expected discounted utility of consumption

$$\max_{\{c_t, a_{t+1}\}} \mathbb{E} \sum_{t=0}^{\infty} \beta^t U(c_t)$$

subject to the following budget constraint:

$$c + a' \leq (1 + r - \delta)a + (1 - \rho)(wz(1 - (1 - \psi)\tau_y)) + \rho b \quad (8)$$

$$a' \geq -\bar{a}$$

where  $b$  is the unemployment benefit.

#### 4.4 Unemployed

The unemployed maximize expected discounted utility of consumption

$$\max_{\{c_t, a_{t+1}\}} \mathbb{E} \sum_{t=0}^{\infty} \beta^t U(c_t)$$

subject to the following budget constraint:

$$c + a' \leq (1 + r - \delta)a + p(\theta) (wz(1 - (1 - \psi)\tau_y)) + (1 - p(\theta))b \quad (9)$$

$$a' \geq -\bar{a}$$

where  $b$  is the unemployment benefit.

#### 4.5 Government

The government runs a balanced budget with a lump-sum taxes  $T(y) = (y - \lambda_p y^{1-\tau_p})$

#### 4.6 The household's problem

Let  $i_w$  denote the household's labor market status in the previous period, with  $i_w = 2$  indicating being entrepreneur,  $i_w = 1$  being employed and  $i_w = 0$  being unemployed. Let  $\mathcal{I}_e$  indicate occupational choice, where  $\mathcal{I}_e = 1$  if the household is an entrepreneur and  $\mathcal{I}_e = 0$  if the household is a worker.

Let denote  $\mathbb{E}_\rho$  the expected value considering the possibility of being separated from the job as employed;  $\mathbb{E}_{p(\theta)}$  the expected value considering the possibility of finding new job if previously unmatched with a firm (unemployed former entrepreneurs).  $V_w$  is the contemporary utility as a worker, which varies with the employment status.  $\iota_{i_w=1}$  is an indicator function. We can write the household's problem recursively as follows. More specifically,

$$\begin{aligned} \mathbb{E}_{p(\theta)} (V_w + \beta \mathbb{E} \mathbf{V}(a', x', i'_w)) &= p(\theta) (V_n + \beta \mathbb{E} \mathbf{V}(a', x', i'_w = 1)) \\ &\quad + (1 - p(\theta)) (V_u + \beta \mathbb{E} \mathbf{V}(a', x', i'_w = 0)) \end{aligned}$$

$$\begin{aligned} \mathbb{E}_\rho (V_w + \beta \mathbb{E} \mathbf{V}(a', x', i'_w)) &= \rho (V_u + \beta \mathbb{E} \mathbf{V}(a', x', i'_w = 0)) \\ &\quad + (1 - \rho) (V_n + \beta \mathbb{E} \mathbf{V}(a', x', i'_w = 1)) \end{aligned}$$



Notice that for agents who recently separated from the firm cannot rematch with a new firm in the current period.

$$\mathbf{V}(a, h, i_w) = \max_{\{a', c, \mathcal{I}_e\}} \left\{ \begin{array}{l} \iota_{i_w=0} [(1 - \mathcal{I}_e) \mathbb{E}_{p(\theta)} (V_w + \beta \mathbb{E} \mathbf{V}(a', x', i'_w)) + \mathcal{I}_e (V_e + \beta \mathbb{E} \mathbf{V}(a', x', i'_w = 0))] + \\ \iota_{i_w=1} [(1 - \mathcal{I}_e) \mathbb{E}_\rho (V_w + \beta \mathbb{E} \mathbf{V}(a', x', i'_w)) + \mathcal{I}_e (V_e + \beta \mathbb{E} \mathbf{V}(a', x', i'_w = 0))] + \\ \iota_{i_w=2} [(1 - \mathcal{I}_e) \mathbb{E}_{p(\theta)} (V_w + \beta \mathbb{E} \mathbf{V}(a', x', i'_w)) + \mathcal{I}_e (V_e + \beta \mathbb{E} \mathbf{V}(a', x', i'_w = 0))] \end{array} \right\}$$

subject to

$$c + a' \leq (1 - \tilde{r} - \delta)a + inc \quad (10)$$

$$inc = \begin{cases} ra + \tilde{w}_n & \text{if } \mathcal{I}_e = 0, i_w = 1 \\ ra + \tilde{w}_u & \text{if } \mathcal{I}_e = 0, i_w \neq 1 \\ ra + \nu(k, x; \tilde{r}, w, \tau) & \text{if } \mathcal{I}_e = 1 \end{cases}$$

where  $inc$  is the earnings of the worker or entrepreneur,  $\tilde{w}_n = (1 - \rho) \{zw(1 - (1 - \psi)\tau)\} + \rho b$  and  $\tilde{w}_u = p(\theta) \{zw(1 - (1 - \psi)\tau)\} + \{1 - p(\theta)\}b$ .

## 4.7 Equilibrium

Let  $\hat{h}$  be the critical value of human capital that determines occupational choice which is affected by both employment status and education level. If  $h \geq \hat{h}$  then individuals choose entrepreneurship over becoming a worker. The critical value of human capital is always higher for those who have employment than those who are unmatched with a firm  $\hat{h}_{i_w \neq 1} < \hat{h}_{i_w = 1}$ . We define necessity entrepreneurship as those who would not have become an entrepreneur *if* they had an employment contract:

$$\hat{h}_{i_w \neq 1} < h < \hat{h}_{i_w = 1}.$$

We define opportunity-driven entrepreneurship as those who would become an entrepreneur *regardless* of circumstance:

$$h \geq \hat{h}_{i_w = 1} \geq \hat{h}_{i_w \neq 1}$$

As there are three education levels, there are a total of six critical values of human capital: two for primary educated matched or unmatched,  $\hat{h}_{i_w = 1}^P$  and  $\hat{h}_{i_w \neq 1}^P$  two for secondary educated,  $\hat{h}_{i_w = 1}^S$  and  $\hat{h}_{i_w \neq 1}^S$ , and two for tertiary educated,  $\hat{h}_{i_w = 1}^T$  and  $\hat{h}_{i_w \neq 1}^T$ .

Table 1: Parameter values, baseline economy

| Parameters    | Values                   | Description                | Comments/observations/target                                       |
|---------------|--------------------------|----------------------------|--------------------------------------------------------------------|
| $\beta$       | 0.94                     | Discount factor            |                                                                    |
| $\sigma$      | 2.0                      | Risk aversion              | $u(c) = \frac{c^{1-\sigma}}{1-\sigma}$                             |
| $\delta$      | 0.06                     | Capital depreciation       |                                                                    |
| $\alpha$      | 0.3207                   | Capital share              |                                                                    |
| $\gamma$      | 0.4693                   | Labour Share               | Frac. of entrepreneurs                                             |
| $h$           | {0.109, 0.581, 0.310}    | Share of population by $h$ | Barro and Lee (2013)                                               |
| $\vartheta_e$ | {0.967, 1.007, 1.156}    | Shape parameter by $h$     | Mean firm size, Gomez and Kuhen (2017)                             |
| $\varsigma$   | 2.02655                  | Location parameter         | $F(x) = \left[1 + \frac{x-\varsigma}{\xi}\right]^{-\alpha_i}$      |
| $\xi$         | 0.1115                   | Scale parameter            | Gomez and Kuhen (2017)                                             |
| $z$           | {1.0, 1.31, 1.61}        | Wage premium by $h$        | Goldin and Katz (2009)                                             |
| $\rho$        | {0.0383, 0.0288, 0.0136} | Job separation rate by $h$ | Cairo and Cajner (2016)                                            |
| $\bar{m}$     | 0.8                      | Matching efficiency        | Lubik, 2013 $p(\theta) = \bar{m} \left(\frac{v}{u}\right)^{1-\mu}$ |
| $\mu$         | [0.5, 0.7]               | Matching elasticity        | Petrongolo and Pissarides (2001)                                   |
| $\kappa$      | [0.03, 0.045] $\cdot w$  | Search costs               | (Hagedorn and Manovskii, 2008)                                     |
| $\tau_\pi$    | 0                        | Corporation tax            | Set as zero for now                                                |
| $\tau_y$      | 0.151                    | Income Tax                 | $T(y) = (y - \lambda_p y^{1-\tau_p})$                              |
| $\lambda_p$   |                          | Income Tax                 | Determined in equm.                                                |
| $b$           | 0.40 $\cdot w$ ,         | Social security benefits   | Shimmer and Krause and Lubik (2005)                                |
| $\psi$        | 0                        | Share                      |                                                                    |
| $\Delta$      | 0                        | Credit spread              |                                                                    |
| $\bar{a}$     | 0                        | Borrowing constraint       |                                                                    |

## 5 Calibration

## 6 Baseline Results

The baseline equilibrium shows moderate entrepreneurship (0.1472 overall), with substantial heterogeneity by education. Entry is highest for high education (0.1618) and lowest for middle education (0.1398), while average labor demand is 21.9129 workers and rises steeply with skill from 9.7237 (low) to 33.5470 (high). This pattern indicates that high-education entrepreneurs operate at much larger scale even when their entry rate is only modestly above other groups. Labor-market transitions also suggest strong persistence in employment (Employed  $\rightarrow$  Work = 0.740) and a small unemployment-to-entrepreneurship channel (0.002), so necessity entry is present but limited in the baseline.

Table 2: Baseline equilibrium outcomes

|                             | Baseline |
|-----------------------------|----------|
| UI                          | 0.40     |
| Interest rate $r$           | 0.0989   |
| Wage $w$                    | 2.3870   |
| Job-finding rate $\theta$   | 0.5063   |
| Income tax $\tau_y$         | 0.0377   |
| Convergence metric          | 0.0010   |
| Govt consumption floor pay  | 332.3780 |
| Entrepreneurship (all edu)  | 0.1472   |
| Entrepreneurship (low edu)  | 0.1457   |
| Entrepreneurship (med edu)  | 0.1398   |
| Entrepreneurship (high edu) | 0.1618   |
| Entr $\rightarrow$ Entr     | 0.145    |
| Entr $\rightarrow$ Work     | 0.000    |
| Emp $\rightarrow$ Entr      | 0.000    |
| Emp $\rightarrow$ Work      | 0.740    |
| Unemp $\rightarrow$ Entr    | 0.002    |
| Unemp $\rightarrow$ Work    | 0.113    |
| Avg labor demand $n$        | 21.9129  |
| Labor demand $n$ (low)      | 9.7237   |
| Labor demand $n$ (med)      | 17.9923  |
| Labor demand $n$ (high)     | 33.5470  |
| Std. dev. labor demand      | 27.9509  |
| Per-capita output           | 4.5316   |
| Output per entrepreneur     | 31.3760  |

## 7 Quantitative Analysis

This section evaluates each policy experiment relative to the baseline and reports how unemployment insurance design and tax closure affect entry, scale, and output.

### 7.1 Low unemployment insurance with endogenous tax

This subsection reports the low-UI endogenous-tax experiment against the baseline.

Table 3: Policy experiment: Baseline vs low unemployment insurance with endogenous tax

|                              | Baseline | Low UI (Endogenous Tax) |
|------------------------------|----------|-------------------------|
| UI                           | 0.40     | 0.20                    |
| Interest rate $r$            | 0.0989   | 0.0982                  |
| Wage $w$                     | 2.3870   | 2.4036                  |
| Job-finding rate $\theta$    | 0.5063   | 0.5137                  |
| Income tax $\tau_y$          | 0.0377   | 0.0182                  |
| Convergence metric           | 0.0010   | 0.0009                  |
| Govt consumption floor pay   | 332.3780 | 218.5260                |
| Entrepreneurship (all edu)   | 0.1472   | 0.1542                  |
| Entrepreneurship (low edu)   | 0.1457   | 0.1542                  |
| Entrepreneurship (med edu)   | 0.1398   | 0.1472                  |
| Entrepreneurship (high edu)  | 0.1618   | 0.1672                  |
| Entr $\rightarrow$ Entr      | 0.145    | 0.151                   |
| Entr $\rightarrow$ Work      | 0.000    | 0.000                   |
| Emp $\rightarrow$ Entr       | 0.000    | 0.000                   |
| Emp $\rightarrow$ Work       | 0.740    | 0.738                   |
| Unemp $\rightarrow$ Entr     | 0.002    | 0.003                   |
| Unemp $\rightarrow$ Work     | 0.113    | 0.107                   |
| Avg labor demand $n$         | 21.9129  | 20.6400                 |
| Labor demand $n$ (low)       | 9.7237   | 9.0788                  |
| Labor demand $n$ (med)       | 17.9923  | 16.8165                 |
| Labor demand $n$ (high)      | 33.5470  | 31.8708                 |
| Std. dev. labor demand       | 27.9509  | 27.1502                 |
| Per-capita output            | 4.5316   | 4.8523                  |
| % vs baseline (pc output)    | 0.00%    | +7.08%                  |
| Output per entrepreneur      | 31.3760  | 31.1302                 |
| % vs baseline (productivity) | 0.00%    | -0.78%                  |

Lower UI increases entrepreneurship and per-capita output relative to baseline, while output per entrepreneur is slightly lower.

## 7.2 No unemployment insurance with endogenous tax

This subsection reports the no-UI endogenous-tax experiment against the baseline.

Table 4: Policy experiment: Baseline vs no unemployment insurance with endogenous tax

|                              | Baseline | No UI (Endogenous Tax) |
|------------------------------|----------|------------------------|
| UI                           | 0.40     | 0.05                   |
| Interest rate $r$            | 0.0989   | 0.0969                 |
| Wage $w$                     | 2.3870   | 2.6273                 |
| Job-finding rate $\theta$    | 0.5063   | 0.4965                 |
| Income tax $\tau_y$          | 0.0377   | 0.0008                 |
| Convergence metric           | 0.0010   | 0.0003                 |
| Govt consumption floor pay   | 332.3780 | 89.6635                |
| Entrepreneurship (all edu)   | 0.1472   | 0.2163                 |
| Entrepreneurship (low edu)   | 0.1457   | 0.2481                 |
| Entrepreneurship (med edu)   | 0.1398   | 0.2180                 |
| Entrepreneurship (high edu)  | 0.1618   | 0.2018                 |
| Entr $\rightarrow$ Entr      | 0.145    | 0.212                  |
| Entr $\rightarrow$ Work      | 0.000    | 0.000                  |
| Emp $\rightarrow$ Entr       | 0.000    | 0.000                  |
| Emp $\rightarrow$ Work       | 0.740    | 0.676                  |
| Unemp $\rightarrow$ Entr     | 0.002    | 0.004                  |
| Unemp $\rightarrow$ Work     | 0.113    | 0.108                  |
| Avg labor demand $n$         | 21.9129  | 11.5534                |
| Labor demand $n$ (low)       | 9.7237   | 4.5716                 |
| Labor demand $n$ (med)       | 17.9923  | 8.7990                 |
| Labor demand $n$ (high)      | 33.5470  | 19.1706                |
| Std. dev. labor demand       | 27.9509  | 18.5538                |
| Per-capita output            | 4.5316   | 4.6991                 |
| % vs baseline (pc output)    | 0.00%    | +3.70%                 |
| Output per entrepreneur      | 31.3760  | 25.1883                |
| % vs baseline (productivity) | 0.00%    | -19.72%                |

Removing UI under endogenous tax strongly raises entry, but average labor demand and output per entrepreneur decline materially.

### 7.3 No unemployment insurance with fixed tax

This subsection reports the no-UI fixed-tax experiment against the baseline.

Table 5: Policy experiment: Baseline vs no unemployment insurance with fixed tax

|                              | Baseline | No UI (Fixed Tax) |
|------------------------------|----------|-------------------|
| UI                           | 0.40     | 0.10              |
| Interest rate $r$            | 0.0989   | 0.0977            |
| Wage $w$                     | 2.3870   | 2.4061            |
| Job-finding rate $\theta$    | 0.5063   | 0.5099            |
| Income tax $\tau_y$          | 0.0377   | 0.0386            |
| Convergence metric           | 0.0010   | 0.0012            |
| Govt consumption floor pay   | 332.3780 | 192.3750          |
| Entrepreneurship (all edu)   | 0.1472   | 0.1530            |
| Entrepreneurship (low edu)   | 0.1457   | 0.1507            |
| Entrepreneurship (med edu)   | 0.1398   | 0.1455            |
| Entrepreneurship (high edu)  | 0.1618   | 0.1680            |
| Entr $\rightarrow$ Entr      | 0.145    | 0.151             |
| Entr $\rightarrow$ Work      | 0.000    | 0.000             |
| Emp $\rightarrow$ Entr       | 0.000    | 0.000             |
| Emp $\rightarrow$ Work       | 0.740    | 0.737             |
| Unemp $\rightarrow$ Entr     | 0.002    | 0.002             |
| Unemp $\rightarrow$ Work     | 0.113    | 0.110             |
| Avg labor demand $n$         | 21.9129  | 20.7240           |
| Labor demand $n$ (low)       | 9.7237   | 9.2638            |
| Labor demand $n$ (med)       | 17.9923  | 16.9940           |
| Labor demand $n$ (high)      | 33.5470  | 31.7442           |
| Std. dev. labor demand       | 27.9509  | 27.1523           |
| Per-capita output            | 4.5316   | 4.5261            |
| % vs baseline (pc output)    | 0.00%    | -0.12%            |
| Output per entrepreneur      | 31.3760  | 27.7709           |
| % vs baseline (productivity) | 0.00%    | -11.49%           |

With fixed tax and no UI, entry rises modestly, per-capita output is roughly unchanged, and productivity per entrepreneur declines.

## 8 Conclusion

This paper develops a high-level quantitative framework for studying necessity entrepreneurship as an equilibrium outcome of occupational choice under labor market frictions. By defining necessity and opportunity entrepreneurship through counterfactual choices, the model separates compositional shifts in entry from changes in aggregate entrepreneurship rates.

The main message is that unemployment risk and social insurance jointly shape who becomes an entrepreneur. Reducing insurance can raise entrepreneurship on paper, but part of this increase reflects necessity entry associated with lower average firm scale. Conversely, stronger insurance can reduce necessity entry and improve sorting, even when effects on total entrepreneurship are modest. These results support evaluating entrepreneurship policy through composition, not levels alone.

Several extensions remain important for the next draft. First, the credit friction wedge

and job-market experiments should be fully completed and integrated into a unified welfare comparison. Second, calibration choices that are still open should be aligned with the code implementation and reported transparently. Third, the literature discussion can be broadened further once the project-specific verified PDF library is restored.

# Appendix

## Computation

Given the values for parameters, the discretized spaces  $\Omega_a$  for  $a$ ,  $\Omega_z$  for  $z$  and  $\Omega_x$  for  $x$ . Each agent draw a human capital level  $h$  (interpreted as educational attainment) at the beginning of the economy, which will be fixed henceforth.  $h$  will determine their labor productivity  $z \in \Omega_z$ , the separation rate  $\rho$ , and managerial ability  $x \in \Omega_x$ . Let  $i_w = 0$  indicating being entrepreneur,  $i_w = 1$  being employed and  $i_w = 2$  being unemployed. The numerical algorithm works as follows.

1. Set a tolerance  $\epsilon > 0$ .
2. Guess  $\Phi^0 = (r^0, w^0, \rho^0, \tau_y^0)$ . Solve for optimal household behavior:

$$f : (\varphi; \Phi) \rightarrow (c, a', \mathcal{I}_e, n, k),$$

where  $\varphi = \{h, z, x, a, i_w\}$ . We will use the method of value function iteration as follows.

- (a) Guess value function  $\mathbf{V}^0(\varphi; \Phi^0)$  and policy functions  $f^0(\varphi; \Phi^0)$ .
- (b) Calculate the cutoff values of  $x_u^*$  (for previously unemployed worker, or entrepreneur) and  $x_n^*$  (for previously employed worker) via the following conditions.

$$\tilde{w}_u = \pi(x_u^*) = \max_n \left\{ (x_u^*)^{1-\alpha-\gamma} k^\alpha n^\gamma - w(1 + \psi\tau)n - rk - \frac{\kappa zn}{q(\theta)} \right\} \quad (11)$$

$$\tilde{w}_n = \pi(x_n^*) = \max_n \left\{ (x_n^*)^{1-\alpha-\gamma} k^\alpha n^\gamma - w(1 + \psi\tau)n - rk - \frac{\kappa zn}{q(\theta)} \right\} \quad (12)$$

where  $\tilde{w}_n$  and  $\tilde{w}_u$  denote the expected compensation for the employed worker and unemployed worker respectively. More specifically,

$$\tilde{w}_n = (1 - \rho) \{zw(1 - (1 - \psi)\tau)\} + \rho b \quad (13)$$

$$\tilde{w}_u = p(\theta) \{zw(1 - (1 - \psi)\tau)\} p(\theta) b. \quad (14)$$

Noting that the separation rate  $\rho$  is pre-determined by the individual's initial human capital level  $h$ .



(c) Update value and policy functions:

$$\mathbf{V}^1(h, z, x, a, i_w) = \max_{\{a', c, \mathcal{I}_e\}} \left\{ \begin{array}{l} \iota_{i_w=0} \cdot \left[ (1 - \mathcal{I}_e) \cdot \mathbb{E}_{p(\theta)} (V_w + \beta \mathbb{E} \mathbf{V}^0(h, z', x', a', i'_w)) \right. \\ \quad \left. + \mathcal{I}_e \cdot (V_e + \beta \mathbb{E} \mathbf{V}^0(h, z', x', a', i'_w = 2)) \right] + \\ \iota_{i_w=1} \cdot \left[ (1 - \mathcal{I}_e) \cdot \mathbb{E}_\rho (V_w + \beta \mathbb{E} \mathbf{V}^0(h, z', x', a', i'_w)) \right. \\ \quad \left. + \mathcal{I}_e \cdot (V_e + \beta \mathbb{E} \mathbf{V}^0(h, z', x', a', i'_w = 2)) \right] + \\ \iota_{i_w=2} \cdot \left[ (1 - \mathcal{I}_e) \cdot \mathbb{E}_{p(\theta)} (V_w + \beta \mathbb{E} \mathbf{V}^0(h, z', x', a', i'_w)) \right. \\ \quad \left. + \mathcal{I}_e \cdot (V_e + \beta \mathbb{E} \mathbf{V}^0(h, z', x', a', i'_w = 2)) \right] \end{array} \right\} \quad (15)$$

$$f^1(\varphi; \Phi^0) = \arg \max \mathbf{V}^1(\varphi; \Phi^0)$$

where  $\mathbb{E}_{p(\theta)} (V_w + \beta \mathbb{E} \mathbf{V}(a', x', i'_w))$  denotes the expected payoff for the individual who is previously unemployed worker, or entrepreneur, and decides to look for paid job, while  $\mathbb{E}_\rho (V_w + \beta \mathbb{E} \mathbf{V}(a', x', i'_w))$  represents the expected payoff for the previously employed worker who decide to stay in workplace.

$$\mathbb{E}_{p(\theta)} (V_w + \beta \mathbb{E} \mathbf{V}(h, z', x', a', i'_w)) = p(\theta) (V_n + \beta \mathbb{E} \mathbf{V}(h, z', x', a', i'_w = 1)) \quad (16)$$

$$+ (1 - p(\theta)) (V_u + \beta \mathbb{E} \mathbf{V}(h, z', x', a', i'_w = 0))$$

$$\mathbb{E}_\rho (V_w + \beta \mathbb{E} \mathbf{V}(h, z', x', a', i'_w)) = (1 - \rho) (V_n + \beta \mathbb{E} \mathbf{V}(h, z', x', a', i'_w = 1)) \quad (17)$$

$$+ \rho (V_u + \beta \mathbb{E} \mathbf{V}(h, z', x', a', i'_w = 0))$$

To simplify the analysis, we assume that the individual who was previously employed and newly separated from the firm is not allowed to search for job at the current period. The measure for the individuals who are looking for jobs,  $u_t$ , the total vacancies  $v_t$ , and job matching rate  $\theta_{n,t}$  are given as follows.

$$u_t = \left( \int_x \iota_{i_w=0} + \int_x \iota_{i_w=2} \right) \cdot \int_0^{x_u^*} \quad (18)$$

$$v_t = \int_x \iota_{i_w=1} \cdot \rho \quad (19)$$

$$\theta_{n,t} = \bar{m} \left( \frac{v_t}{u_t} \right)^{1-\mu} \quad (20)$$

Note that we can write the above value functions explicitly as a system of sub-problems:  $P_{x < x_u^*}^{i_w=0}$ ,  $P_{x \geq x_u^*}^{i_w=0}$ ,  $P_{x < x_n^*}^{i_w=1}$ ,  $P_{x \geq x_n^*}^{i_w=1}$ ,  $P_{x < x_u^*}^{i_w=2}$ ,  $P_{x \geq x_u^*}^{i_w=2}$ , which correspond to the dynamic programming problem for i). previously self-employed with  $x < x_u^*$ ; ii). previously self-employed with  $x \geq x_u^*$ ; iii). previously employed with  $x < x_n^*$ ; iv). previously employed with  $x \geq x_n^*$ ; v). previously unemployed with  $x < x_u^*$ ; vi). previously unemployed with  $x \geq x_u^*$ . To economize the space, we only provide the details for

$P_{x < x_u^*}^{i_w=0}$ , and  $P_{x \geq x_u^*}^{i_w=0}$  as follows.  
 $\left(P_{x < x_u^*}^{i_w=0}\right)$ :

$$\mathbf{V}^1(h, z, x < x_u^*, a, i_w = 0) = \max_{a'_n, a'_u} \left\{ \begin{array}{l} p(\theta) (u(c_n) + \beta \mathbb{E} \mathbf{V}^0(h, z', x', a'_n, i'_w = 1)) \\ + (1 - p(\theta)) (u(c_u) + \beta \mathbb{E} \mathbf{V}^0(h, z', x', a'_u, i'_w = 0)) \end{array} \right\} \quad (21)$$

s.t.

$$c_n + a'_n \leq (1 - \tilde{r} - \delta)a + zw(1 - (1 - \psi)\tau), \text{ if matched with a firm} \quad (22)$$

$$c_u + a'_u \leq (1 - \tilde{r} - \delta)a + b, \text{ if unemployed} \quad (23)$$

$\left(P_{x \geq x_u^*}^{i_w=0}\right)$ :

$$\mathbf{V}^1(h, z, x \geq x_u^*, a, i_w = 0) = \max_{a'_e} \{u(c_e) + \beta \mathbb{E} \mathbf{V}^0(h, z', x', a'_e, i'_w = w)\} \quad (24)$$

s.t.

$$c_e + a'_e \leq (1 - \tilde{r} - \delta)a + \nu(k, x; \tilde{r}, w, \tau) \quad (25)$$

where

$$\nu(k, x; \tilde{r}, w, \tau) = \max_n \left\{ x^{1-\alpha-\gamma} k^\alpha n^\gamma - w(1 + \psi\tau)n - rk - \frac{\kappa \rho n}{q(\theta)} \right\}. \quad (26)$$

(d) Stop if  $\max \{|\mathbf{V}^1 - \mathbf{V}^0|, |f^1 - f^0|\} \leq \epsilon$ . Set  $\mathbf{V}^* = \mathbf{V}^1$ , and  $f^* = f^1$ . Otherwise, set  $\mathbf{V}^0 = \mathbf{V}^1$ ,  $f^0 = f^1$  and repeat from step (c).

3. Generate a large number of individuals,  $N = 100000$ . For each agent  $j$  assign a vector of initial condition  $(h_t^j, z_0^j, x_0^j, a_0^j, i_{w,0}^j)$ , where  $x_0^j \sim \Omega_x$ ,  $z_0^j \in \Omega_z$ ,  $i_{w,0}^j \in \Omega_w$ .
4. Simulate the economy for  $T$  periods, where  $T$  is sufficiently large.
5. Calculate the following statistics from the simulated path

$$\left\{ h_t^j, z_t^j, x_t^j, a_t^j, i_{w,t}^j, \mathcal{I}_{e,t}^j, n_t^j, k_t^j \right\}_{t=0}^T.$$

$$LS^0 = \frac{\sum_{j=1}^N (\iota_{i_w=1} \cdot z^j)}{N} \quad (27)$$

$$KS^0 = \frac{\sum_{j=1}^N (\iota_{i_w=0} \cdot a^j + \iota_{i_w=1} \cdot a^j + \iota_{i_w=2} \cdot a^j - \iota_{i_w=0} \cdot k^j)}{\sum_{j=1}^N a^j} \quad (28)$$

$$u_t = \left( \int_h \iota_{i_w=0} + \int_h \iota_{i_w=2} \right) \cdot \int_0^{h_u^*} \quad (29)$$

$$v_t = \int_h \iota_{i_w=1} \cdot \rho \quad (30)$$

$$p(\theta)_t = \bar{m} \left( \frac{v_t}{u_t} \right)^{1-\mu} \quad (31)$$

$$G = \int \tau w z \quad (32)$$

and  $\tau_y^1$  that balances the government's budget.

6. Stop and set  $(r^*, w^*, \theta^*) = (r^0, w^0, \theta^0)$ , if  $\max \{LS^0, KS^0, |\theta_n^1 - \theta_n^0|\} \leq \epsilon$ . Otherwise, update aggregate variables (restart from step 2):

$$\begin{aligned} r^0 &= \chi r^0 + (1 - \chi) r^1 (KS^0) \\ w^0 &= \chi w^0 + (1 - \chi) w^1 (LS^0) \\ p(\theta)^0 &= \chi p(\theta)^0 + (1 - \chi) p(\theta)^1 \\ \tau_y^0 &= \chi \tau_y^0 + (1 - \chi) \tau_y^1 \end{aligned}$$

where  $\chi \in (0, 1)$  is the step for updating aggregate variables.

## Appendix Search and Matching

I am not sure how much of this is needed for coding purposes but this is the standard search and matching model. The stock of employed workers  $n_t$  can be expressed as the current pool of workers who did not separate from firms, plus new hires

$$n_t = (1 - \rho)n_{t-1} + m_{t-1}$$

The matching function is Cobb-Douglass where

$$m_t = \bar{m} v_t^{1-\mu} u_t^\mu$$

$\bar{m}$  is a match efficiency parameter,  $v$  is vacancies,  $u$  is unemployment, and match elasticity is  $0 < \mu < 1$ . The hiring rate can be defined for firms as total number of new hires to the number of vacancies

$$q(\theta_t) = \frac{m_t}{v_t} = \bar{m}\theta_t^{-\mu}$$

where  $\theta_t = \frac{v_t}{u_t}$  is aggregate labour market tightness. Similarly we can define this from the perspective of the worker in the job-finding rate.

$$p(\theta_t) = \frac{m_t}{u_t} = \bar{m}\theta_t^{1-\mu}$$

The Job Creation Condition determines the amount of new vacancies an employer will post. In our model, most workers are on rolling contracts, so firms only need to post new vacancies after some workers separate from these rolling contracts. Usually in search and matching models the wage is determined ex-post of matching by Nash bargaining between the employer and employee. However, in our span-of-control model the wage has already been determined ex ante, as the wage is an endogenous variable that determines occupational choice. The marginal benefit that the new employee would bring to the firm is expressed as  $A$  (which can be derived from the firms production function,  $\gamma X k^\alpha (zn)^{\gamma-1}$ ). The *JCC* can be then expressed as the costs of replacing out workers on the *LHS* and the expected benefit on the *RHS*

$$\frac{\kappa}{q(\theta_t)} = A - w_t + \beta E_t \left[ A - w_{t+1} + (1 - \rho) \frac{\kappa}{q(\theta_{t+1})} \right]$$

where  $\kappa$  is the cost of posting a vacancy.

## Matching Steady States

To find the steady state of the employment equation  $n_t = (1 - \rho)n_{t-1} + m_{t-1}$ , we use the fact that  $u = (1 - n)$  and  $m_t = \bar{m}v_t^{1-\mu}u_t^\mu$ . Substituting and getting rid of time subscript gives  $(1 - u) = (1 - \rho)(1 - u) + \bar{m}v^{1-\mu}u^\mu$  which simplifies to  $\bar{m}v^{1-\mu}u^\mu = \rho(1 - u)$ . Making  $v$  the subject of the formula gives us the Beveridge curve.

$$v = \left( \frac{\rho(1 - u)}{\bar{m}u} \right)^{1-\mu} u$$

The steady state for the JCC can be found by using  $\frac{\kappa}{q(\theta_t)} = A - w_t + \beta E_t \left[ A - w_{t+1} + (1 - \rho) \frac{\kappa}{q(\theta_{t+1})} \right]$  and  $q(\theta_t) = \bar{m}\theta_t^{-\mu} = \bar{m} \left( \frac{v_t}{u_t} \right)^{-\mu}$ . Substituting and getting rid of time subscripts yields

$$\frac{\kappa}{\bar{m}} \left( \frac{v}{u} \right)^\mu = A - w_t + \beta \left[ A - w + (1 - \rho) \frac{\kappa}{\bar{m}} \left( \frac{v}{u} \right)^\mu \right]$$

Simplifying yields the steady state of the JCC

$$1 - \beta(1 - \rho) \frac{\kappa}{\bar{m}} \left( \frac{v}{u} \right)^\mu = (1 + \beta) [A - w]$$

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